Methodology

1. Governing Equations

$$M(q)\,\ddot{q} + R(q, \dot{q}, t) = 0 \tag{1}$$

$$g(q) = 0 \tag{2}$$

Description: System is modeled using nonlinear ODEs with geometric constraints.

2. KKT Formulation

$$M(q)\,\ddot{q} + R(q, \dot{q}, t) + G(q)^T \lambda = 0 \tag{3}$$

$$g(q) = 0 \tag{4}$$

Description: Constraints enforced using Lagrange multipliers. $G(q) = \frac{\partial g}{\partial q}$ is the constraint Jacobian.

3. Constraint Differentiation

$$\dot{g}(q) = G(q)\,\dot{q} = 0 \tag{5}$$

$$\ddot{g}(q) = G(q)\,\ddot{q} + \dot{G}(q)\,\dot{q} = 0 \tag{6}$$

Description: Provides velocity and acceleration-level constraint equations.

4. KKT with Baumgarte Stabilization

$$M(q)\,\ddot{q} + R(q, \dot{q}, t) + G(q)^T \lambda = 0 \tag{7}$$

Baumgarte Stabilization:

$$\gamma = 2\zeta\omega\,\dot{g}(q) + \omega^2 g(q) \tag{8}$$

$$G(q)\,\ddot{q} = -\gamma \tag{9}$$

Description: Baumgarte stabilization modifies the acceleration-level constraint by adding damping and stiffness terms, reducing drift.

5. Final System (Matrix Form)

$$\begin{bmatrix} M(q) & G(q)^T \\ G(q) & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} -R(q, \dot{q}, t) \\ -\gamma \end{bmatrix} \tag{10}$$

Description: Linear system solved to obtain accelerations and constraint forces.

6. Highlighted Final Equation

$$\boxed{\begin{bmatrix} M & G^T \\ G & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} -R \\ -\gamma \end{bmatrix}} \tag{11}$$

Objective:

$$\min J = \mathrm{RMS}(a_{\mathrm{cabin}})$$

Design Variables:

$$K_F,\ C_F,\ K_2,\ K_3$$

Constraints:

$$K_i^{\min} \leq K_i \leq K_i^{\max}, \quad C_i^{\min} \leq C_i \leq C_i^{\max}$$

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\mathrm{phy}} + \lambda_2 \mathcal{L}_{\mathrm{data}} + \lambda_3 \mathcal{L}_{\mathrm{sm}} + \lambda_4 \mathcal{L}_{\mathrm{IC}} + \lambda_5 \mathcal{L}_{\mathrm{vel}} + \lambda_6 \mathcal{L}_{\mathrm{acc}} + \lambda_7 \mathcal{L}_{\mathrm{geom\_const}}$$

Metric	**Value**
RMSE	7.28×10^{-3}
MAE	4.93×10^{-3}
R^2	0.716
Accuracy	72.1%